

Creation Date 12-Dec-1997

Revision Date 09-Feb-2024

Revision Number 9

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: Titanium(III) Sulfate, ca 15% solution in diluted sulfuric acid  
Cat No. : T/2000/PB08

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

For customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Substances/mixtures corrosive to metal

Category 1 (H290)

### Health hazards

Skin Corrosion/Irritation

Category 2 (H315)

Serious Eye Damage/Eye Irritation

Category 2 (H319)

### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Warning

### Hazard Statements

H290 - May be corrosive to metals

H315 - Causes skin irritation

H319 - Causes serious eye irritation

### Precautionary Statements

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P390 - Absorb spillage to prevent material damage

## 2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component                              | CAS No     | EC No             | Weight %   | CLP Classification - Regulation (EC) No 1272/2008 |
|----------------------------------------|------------|-------------------|------------|---------------------------------------------------|
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | EEC No. 233-749-9 | >=10 - <15 | -                                                 |
| Sulfuric acid                          | 7664-93-9  | 231-639-5         | >=10 - <15 | Skin Corr. 1A (H314)<br>Eye Dam. 1 (H318)         |
| Water                                  | 7732-18-5  | 231-791-2         | 70 - 80    | -                                                 |

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| Component     | Specific concentration limits (SCL's)                                              | M-Factor | Component notes |
|---------------|------------------------------------------------------------------------------------|----------|-----------------|
| Sulfuric acid | Skin Corr. 1A :: C>=15%<br>Eye Irrit. 2 :: 5%<=C<15%<br>Skin Irrit. 2 :: 5%<=C<15% | -        | -               |

| Components    | Reach Registration Number |
|---------------|---------------------------|
| Sulfuric acid | 01-2119458838-20          |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                        |
|---------------------------|------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. |
|---------------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

#### Hazardous Combustion Products

Thermal decomposition can lead to release of irritating gases and vapors, Sulfur oxides, Hydrogen.

### 5.3. Advice for firefighters

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As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation.  
Refer to protective measures listed in Sections 7 and 8

### 6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 12

**Switzerland - Storage of hazardous substances**

Storage class - SC 10/12  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### **Exposure limits**

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

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| Component     | European Union                                                                           | The United Kingdom                                                                                                                                                                                                                                          | France                                                                                                                        | Belgium                                                                                       | Spain                                                                                                                                           |
|---------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Sulfuric acid | TWA: 0.05 mg/m <sup>3</sup> (8h)                                                         | STEL: 0.15 mg/m <sup>3</sup> 15 min<br>TWA: 0.05 mg/m <sup>3</sup> 8 hr                                                                                                                                                                                     | TWA / VME: 0.05 mg/m <sup>3</sup> (8 heures).                                                                                 | TWA: 0.2 mg/m <sup>3</sup> 8 uren                                                             | TWA / VLA-ED: 0.05 mg/m <sup>3</sup> (8 horas)                                                                                                  |
| Component     | Italy                                                                                    | Germany                                                                                                                                                                                                                                                     | Portugal                                                                                                                      | The Netherlands                                                                               | Finland                                                                                                                                         |
| Sulfuric acid | -                                                                                        | TWA: 0.1 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>TWA: 0.1 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.1 mg/m <sup>3</sup>                                                                                                          | TWA: 0.2 mg/m <sup>3</sup> 8 horas                                                                                            | TWA: 0.05 mg/m <sup>3</sup> 8 uren                                                            | TWA: 0.05 mg/m <sup>3</sup> 8 tunteina<br>STEL: 0.1 mg/m <sup>3</sup> 15 minuutteina                                                            |
| Component     | Austria                                                                                  | Denmark                                                                                                                                                                                                                                                     | Switzerland                                                                                                                   | Poland                                                                                        | Norway                                                                                                                                          |
| Sulfuric acid | MAK-KZGW: 0.2 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 0.1 mg/m <sup>3</sup> 8 Stunden   | TWA: 0.05 mg/m <sup>3</sup> 8 timer<br>STEL: 0.1 mg/m <sup>3</sup> 15 minutter                                                                                                                                                                              | STEL: 0.2 mg/m <sup>3</sup> 15 Minuten<br>TWA: 0.1 mg/m <sup>3</sup> 8 Stunden                                                | TWA: 0.05 mg/m <sup>3</sup> 8 godzinach                                                       | TWA: 0.1 mg/m <sup>3</sup> 8 timer<br>STEL: 0.3 mg/m <sup>3</sup> 15 minutter. value calculated thoracic fraction, aerosol                      |
| Component     | Bulgaria                                                                                 | Croatia                                                                                                                                                                                                                                                     | Ireland                                                                                                                       | Cyprus                                                                                        | Czech Republic                                                                                                                                  |
| Sulfuric acid | TWA: 0.05 mg/m <sup>3</sup>                                                              | TWA-GVI: 0.05 mg/m <sup>3</sup> 8 satima. when selecting the appropriate exposure monitoring method the potential limitations and disturbances that may occur in the presence of other sulfur compounds should be taken into account fog, thoracic fraction | TWA: 0.05 ppm 8 hr.<br>STEL: 0.15 ppm 15 min                                                                                  | TWA: 0.05 mg/m <sup>3</sup>                                                                   | TWA: 1 mg/m <sup>3</sup> 8 hodinách. SO3<br>TWA: 0.05 mg/m <sup>3</sup> 8 hodinách. concentrated H2SO4 mist<br>Ceiling: 2 mg/m <sup>3</sup> SO3 |
| Component     | Estonia                                                                                  | Gibraltar                                                                                                                                                                                                                                                   | Greece                                                                                                                        | Hungary                                                                                       | Iceland                                                                                                                                         |
| Sulfuric acid | TWA: 0.05 mg/m <sup>3</sup> 8 tundides. particles that reach the upper respiratory tract | TWA: 0.05 mg/m <sup>3</sup> 8 hr when selecting an appropriate exposure monitoring method, account should be taken of potential limitations and interferences that may arise in the presence of other sulphur compounds thoracic fraction                   | TWA: 0.05 mg/m <sup>3</sup>                                                                                                   | TWA: 0.05 mg/m <sup>3</sup> 8 órában. AK                                                      | TWA: 0.05 mg/m <sup>3</sup> 8 klukkustundum. thoracic fraction<br>Ceiling: 0.1 mg/m <sup>3</sup> aerosol                                        |
| Component     | Latvia                                                                                   | Lithuania                                                                                                                                                                                                                                                   | Luxembourg                                                                                                                    | Malta                                                                                         | Romania                                                                                                                                         |
| Sulfuric acid | TWA: 0.05 mg/m <sup>3</sup>                                                              | TWA: 0.05 mg/m <sup>3</sup> vapor IPRD<br>STEL: 3 mg/m <sup>3</sup>                                                                                                                                                                                         | TWA: 0.05 mg/m <sup>3</sup> 8 Stunden                                                                                         | TWA: 0.05 mg/m <sup>3</sup>                                                                   | TWA: 0.05 mg/m <sup>3</sup> 8 ore                                                                                                               |
| Component     | Russia                                                                                   | Slovak Republic                                                                                                                                                                                                                                             | Slovenia                                                                                                                      | Sweden                                                                                        | Turkey                                                                                                                                          |
| Sulfuric acid | Skin notation<br>MAC: 1 mg/m <sup>3</sup>                                                | TWA: 0.05 mg/m <sup>3</sup>                                                                                                                                                                                                                                 | TWA: 0.05 mg/m <sup>3</sup> 8 urah inhalable fraction, fog<br>STEL: 0.05 mg/m <sup>3</sup> 15 minutah inhalable fraction, fog | Indicative STEL: 0.2 mg/m <sup>3</sup> 15 minuter<br>TLV: 0.1 mg/m <sup>3</sup> 8 timmar. NGV | TWA: 0.05 mg/m <sup>3</sup> 8 saat                                                                                                              |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

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BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                 | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-------------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Sulfuric acid<br>7664-93-9 ( >=10 - <15 ) | DNEL = 0.1mg/m <sup>3</sup>      |                                     | DNEL = 0.05mg/m <sup>3</sup>       |                                       |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                                 | Fresh water       | Fresh water sediment          | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture) |
|-------------------------------------------|-------------------|-------------------------------|--------------------|------------------------------------|--------------------|
| Sulfuric acid<br>7664-93-9 ( >=10 - <15 ) | PNEC = 0.0025mg/L | PNEC = 0.002mg/kg sediment dw |                    | PNEC = 8.8mg/L                     |                    |

| Component                                 | Marine water       | Marine water sediment         | Marine water Intermittent | Food chain | Air |
|-------------------------------------------|--------------------|-------------------------------|---------------------------|------------|-----|
| Sulfuric acid<br>7664-93-9 ( >=10 - <15 ) | PNEC = 0.00025mg/L | PNEC = 0.002mg/kg sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-----------------------------------------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber<br>Nitrile rubber<br>Neoprene<br>PVC | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

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**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | Purple                   |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flammability (liquid)</b>                   | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | Not applicable           | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>pH</b>                                      | < 1                      |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | Miscible                 |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | 1.28                     |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

### 9.2. Other information

## SECTION 10: STABILITY AND REACTIVITY

**10.1. Reactivity** None known, based on information available

**10.2. Chemical stability** Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

**Hazardous Polymerization** No information available.  
**Hazardous Reactions** None under normal processing.

**10.4. Conditions to avoid** Incompatible products. Excess heat.

**10.5. Incompatible materials** Bases. Metals.

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## 10.6. Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapors. Sulfur oxides. Hydrogen.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

No data available

Inhalation

No data available

#### Toxicology data for the components

| Component     | LD50 Oral          | LD50 Dermal | LC50 Inhalation               |
|---------------|--------------------|-------------|-------------------------------|
| Sulfuric acid | 2140 mg/kg ( Rat ) | -           | LC50 = 0.375 mg/L ( Rat ) 4 h |
| Water         | -                  | -           | -                             |

##### (b) skin corrosion/irritation;

Category 2

##### (c) serious eye damage/irritation;

Category 2

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

##### (e) germ cell mutagenicity;

No data available

##### (f) carcinogenicity;

No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component     | EU | UK | Germany | IARC    |
|---------------|----|----|---------|---------|
| Sulfuric acid |    |    |         | Group 1 |

##### (g) reproductive toxicity;

No data available

##### (h) STOT-single exposure;

No data available

##### (i) STOT-repeated exposure;

No data available

Target Organs

No information available.

##### (j) aspiration hazard;

No data available

Symptoms / effects, both acute and delayed

No information available.

### 11.2. Information on other hazards

#### Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any



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known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects

| Component     | Freshwater Fish                                     | Water Flea        | Freshwater Algae |
|---------------|-----------------------------------------------------|-------------------|------------------|
| Sulfuric acid | LC50: > 500 mg/L, 96h static<br>(Brachydanio rerio) | EC50: 29 mg/L/24h | -                |

| Component     | Microtox | M-Factor |
|---------------|----------|----------|
| Sulfuric acid | -        |          |

### 12.2. Persistence and degradability

Persistence

Miscible with water, Persistence is unlikely, based on information available.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

Persistent Organic Pollutant

This product does not contain any known or suspected substance

Ozone Depletion Potential

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Empty containers should be taken to an approved waste handling site for recycling or disposal. Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Solutions with low pH-value must be neutralized before discharge.

Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

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## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

**14.1. UN number** UN1760  
**14.2. UN proper shipping name** Corrosive liquid, n.o.s.  
**Technical Shipping Name** Contains Sulfuric acid  
**14.3. Transport hazard class(es)** 8  
**14.4. Packing group** III

### ADR

**14.1. UN number** UN1760  
**14.2. UN proper shipping name** Corrosive liquid, n.o.s.  
**Technical Shipping Name** Contains Sulfuric acid  
**14.3. Transport hazard class(es)** 8  
**14.4. Packing group** III

### IATA

**14.1. UN number** UN1760  
**14.2. UN proper shipping name** Corrosive liquid, n.o.s.  
**Technical Shipping Name** Contains Sulfuric acid  
**14.3. Transport hazard class(es)** 8  
**14.4. Packing group** III

**14.5. Environmental hazards** No hazards identified  
**14.6. Special precautions for user** No special precautions required.  
**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                              | CAS No     | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|----------------------------------------|------------|-----------|--------|-----|-------|------|----------|------|------|
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | 233-749-9 | -      | -   | X     | X    | KE-12744 | -    | X    |
| Sulfuric acid                          | 7664-93-9  | 231-639-5 | -      | -   | X     | X    | KE-32570 | X    | X    |
| Water                                  | 7732-18-5  | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |

| Component                              | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|----------------------------------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | X    | ACTIVE                                        | -   | X    | X    | X     | -     |
| Sulfuric acid                          | 7664-93-9  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Water                                  | 7732-18-5  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component | CAS No | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous | REACH Regulation (EC 1907/2006) article 59 - Candidate List of |
|-----------|--------|---------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------|
|-----------|--------|---------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------|

# SAFETY DATA SHEET

Titanium(III) Sulfate, ca 15% solution in diluted sulfuric acid

Revision Date 09-Feb-2024

|                                        |            |   | Substances                                                         | Substances of Very High Concern (SVHC) |
|----------------------------------------|------------|---|--------------------------------------------------------------------|----------------------------------------|
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | - | -                                                                  | -                                      |
| Sulfuric acid                          | 7664-93-9  | - | Use restricted. See item 75.<br>(see link for restriction details) | -                                      |
| Water                                  | 7732-18-5  | - | -                                                                  | -                                      |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component                              | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | Not applicable                                                                            | Not applicable                                                                           |
| Sulfuric acid                          | 7664-93-9  | Not applicable                                                                            | Not applicable                                                                           |
| Water                                  | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 1 (self classification)

| Component     | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|---------------|---------------------------------------|-------------------------|
| Sulfuric acid | WGK1                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                                 | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Sulfuric acid<br>7664-93-9 ( >=10 - <15 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

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## Full text of H-Statements referred to under sections 2 and 3

H290 - May be corrosive to metals  
H314 - Causes severe skin burns and eye damage  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H319 - Causes serious eye irritation

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## **Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## **Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Creation Date** 12-Dec-1997

**Revision Date** 09-Feb-2024

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information

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relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**